

TEST REPORT

Report No.: **CHTEW21100067** Report verification : 

Project No.: **SHT2107068302EW**

Applicant's name: **Chengdu Ebyte Electronic Technology Co., Ltd.**

Address: Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China

Test item description: **SPI SMD LoRa Module**

Trade Mark: EBYTE

Model/Type reference: E22-900M22S

Listed Model(s): -

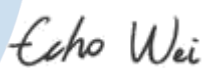
Standard: **EN IEC 62311:2020**

Date of receipt of test sample: Jul.26, 2021

Date of testing: Jul.26, 2021- Oct.14, 2021

Date of issue: Oct.15, 2021

Result: **PASS**

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Testing Laboratory Name: **Shenzhen Huatongwei International Inspection Co., Ltd.**

Address: 1/F, Bldg 3, Hongfa Hi-tech Industrial Park, Genyu Road, Tianliao, Gongming, Shenzhen, China

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The test report merely correspond to the test sample.

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1. Test standards and Report version

1.1. Test standards

The tests were performed according to following standards:

[EN IEC 62311:2020](#)- Assessment of electronic and electrical equipment related to human exposure restrictions for electromagnetic fields (0 Hz – 300 GHz) .

[EN62232:2017](#)- Determination of RF field strength, power density and SAR in the vicinity of radio communication base stations for the purpose of evaluating human exposure.

1.2. Report version information

Revision No.	Date of issue	Description
N/A	2021-10-15	Original

2. Summary

2.1. Client Information

Applicant:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China
Manufacturer:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China
Factory:	Chengdu Ebyte Electronic Technology Co., Ltd.
Address:	Building B5, Mould Industrial Park, 199# Xiqu Ave, West High-tech Zone, Chengdu, China

2.2. Product Description

Name of EUT:	SPI SMD LoRa Module		
Trade Mark:	EBYTE		
Model/Type reference:	E22-900M22S		
Listed Model(s):	-		
Power supply:	DC 5.0V		
Hardware version:	V1.0		
Software version:	V1.0		
Transmitter unit:			
Operation Frequency:	863MHz~870MHz		
Modulation type:	LORA		
Modulation connector:	☒ Without external ☐ External		
Occupied bandwidth ^{#1} :	125KHz		
Product type:	☒ Wideband Tranceiver ☐ Narrowband Tranceiver		
Antenna type:	Sucker antenna		
Antenna gain ^{#2} :	3dBi		
Receiver unit:			
Receiver category ^{#3} :	☒ 2	☐ 1.5	☐ 1

Note:

#1: Declared by the manufacturer

#2: The antenna gain is provided by the applicant, and the applicant should be responsible for its authenticity, HTW lab has not verified the authenticity of its information.

#3: Declared by the manufacturer

2.3. Testing Laboratory Information

Laboratory Name	Shenzhen Huatongwei International Inspection Co., Ltd.
Laboratory Location	1/F, Bldg 3, Hongfa Hi-tech Industrial Park, Genyu Road, Tianliao, Gongming, Shenzhen, China
Connect information:	Tel: 86-755-26715499 E-mail: cs@szhtw.com.cn http://www.szhtw.com.cn

3. RF Exposure

LIMIT

Reference levels for electric, magnetic and electromagnetic fields (0Hz to 300GHz, unperturbed rms values)

Frequency range	E-field strength (V/m)	H-field strength (A/m)	B-field (uT)	Equivalent plane wave power density Seq(W/m2)
0-1Hz	--	3.2×10^4	4×10^4	--
1-8Hz	10000	$3.2 \times 10^4 / f^2$	$4 \times 10^4 / f^2$	--
8-25Hz	10000	$4000 / f$	$5000 / f$	--
0.025-0.8KHz	$250 / f$	$4 / f$	$5 / f$	--
0.8-3KHz	$250 / f$	5	6.25	--
3-150KHz	87	5	6.25	--
0.15-1MHz	87	$0.73 / f$	$0.92 / f$	--
1-10MHz	$87 f^{1/2}$	$0.73 / f$	$0.92 / f$	--
10-400MHz	28	0.073	0.092	2
400-2000MHz	$1.375 f^{1/2}$	$0.0037 f^{1/2}$	$0.0046 f^{1/2}$	$f / 200$
2-300GHz	61	0.16	0.20	10

Notes:

- As indicated in the frequency range column.
- For frequencies between 100kHz and 10GHz, S_{eq} , E^2 , H^2 and B^2 are to be averaged over any six-minute period.
- For frequencies exceeding 10GHz, S_{eq} , E^2 , H^2 and B^2 are to be averaged over any $68/f^{1.05}$ -minute period (.in GHz).
- No E-field value is provided for frequencies <1Hz, which are effectively static electric fields. For most people the annoying perception of surface electric charges will not occur at field strengths less than 20kV/m. Spark discharges causing stress or annoyance should be avoided.

MPE Calculation Method

Equation from page 98 of EN 62232, Edition 2017

$$E = \eta_0 H = \frac{\sqrt{30PG(\theta, \phi)}}{r}$$

Where:

E: E-field strength (V/m)

P: power input to antenna (Watt)

G: is the antenna gain relative to an isotropic antenna;

θ, ϕ : are elevation and azimuth angles to point of investigation;

r: is the distance from observation point to the antenna;

η_0 : is the characteristic impedance of free space.

TEST RESULTS

☒ Passed ☐ Not Applicable

Type	Test channel	Conducted output power (dBm)	r (m)	E-field strength (V/m)	Limit (V/m)@ E-field strength	Result
LORA	CH-L	9.17	0.2	3.5199	61	Pass
	CH-M	8.48	0.2	3.2519		
	CH-H	7.96	0.2	3.0619		

Note:

1. The max antenna gain is 3dBi
2. r is the distance from observation point to the antenna which is declared by the applicant.

-----End of Report-----